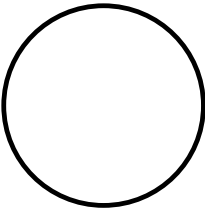
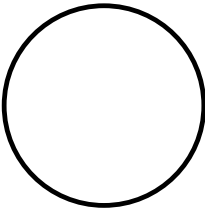
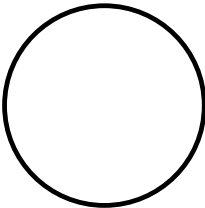
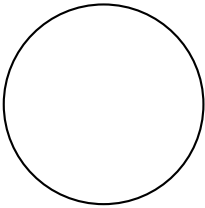
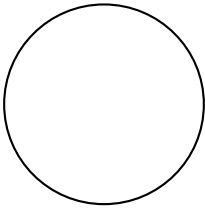
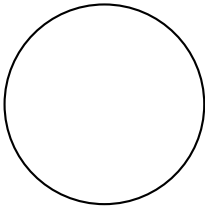
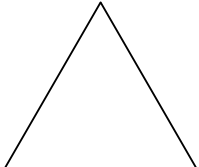
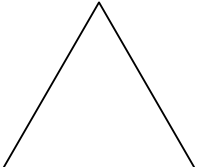
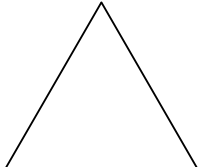


In the code game, the keeper hides a secret row of counters, each red or yellow, behind the folder. The breaker builds a test in the next test row, with a red or yellow counter in every slot. The keeper copies the test just below the secret, puts a green triangle under each place where the two colors match, and says how many triangles there are. That number is the test's score, and the breaker writes it in the score box. When the breaker says what the secret is, the keeper lifts the folder and every score is checked again. A third player can be the referee, who watches every score being made.

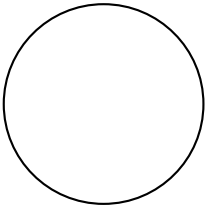
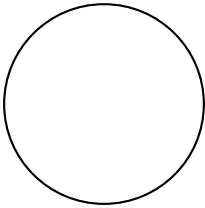
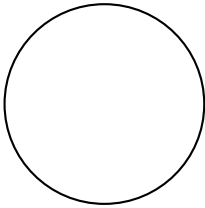
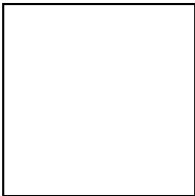
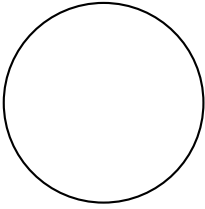
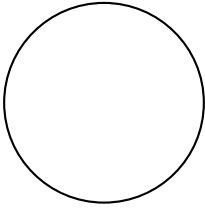
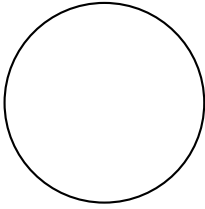
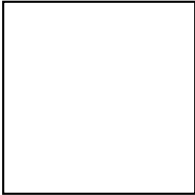
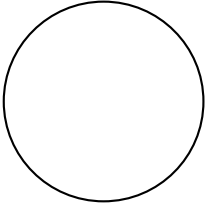
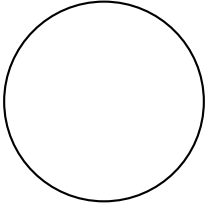
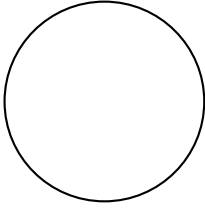
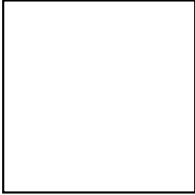
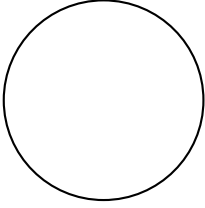
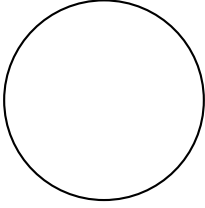
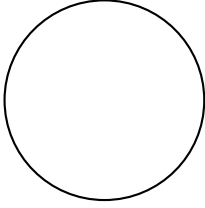
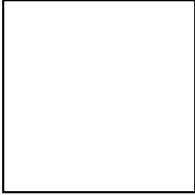
Problem 1: Your partner thinks of a whole number from 1 to 16. Ask yes-or-no questions until you are sure of the number. What is the smallest number of questions that always works? Explain why fewer questions cannot always work. Then answer the same questions for a number from 1 to 20.

[illegible]

Problem 2: Play the code game with three counters, taking turns. Find each secret with as few tests as you can.

secret			
copy of test			
			

----- folder

test				score 
				
				
				

Problem 3: For each of these games, find every secret that fits all the scores. If no secret fits, explain how that could happen.

			score
R	R	Y	2
R	Y	R	2
Y	R	R	2

			score
R	R	R	2
R	Y	Y	2

				score
R	R	Y	Y	3
Y	R	Y	R	1
R	R	R	Y	2

				score
R	R	Y	Y	2
R	Y	R	Y	2
R	Y	Y	R	2

				score
R	Y	R	Y	1
R	R	Y	Y	3
Y	Y	Y	Y	2

Problem 4: Find a way to choose your tests so that you always know a three-counter secret after three tests or fewer. Describe it so that someone else could use it.

			score

			score

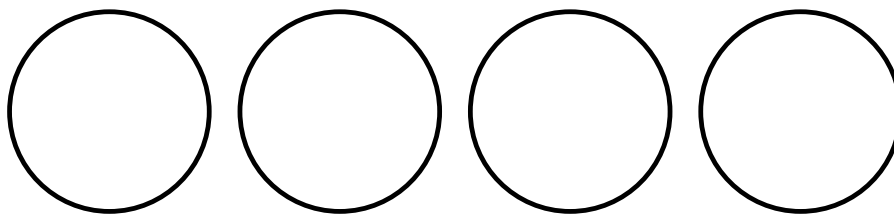
			score

			score

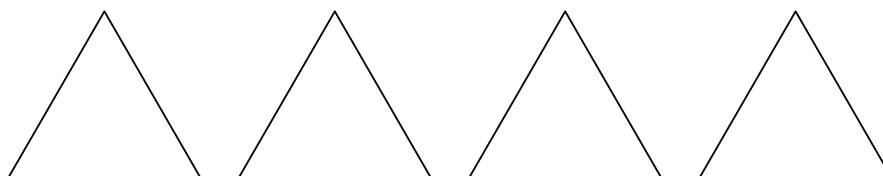
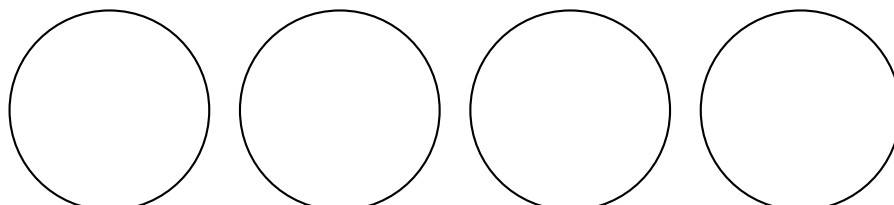
This image shows a single sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins or other markings on the paper.

Problem 5: Play the code game with four counters, taking turns. Find each secret with as few tests as you can.

secret

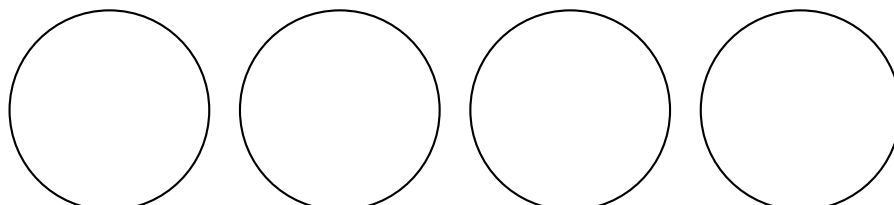
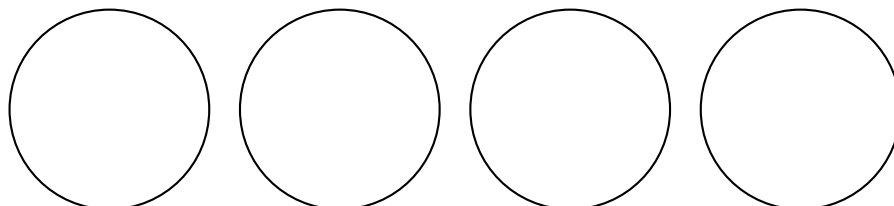
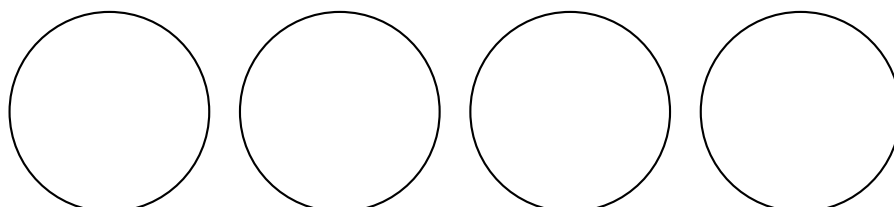
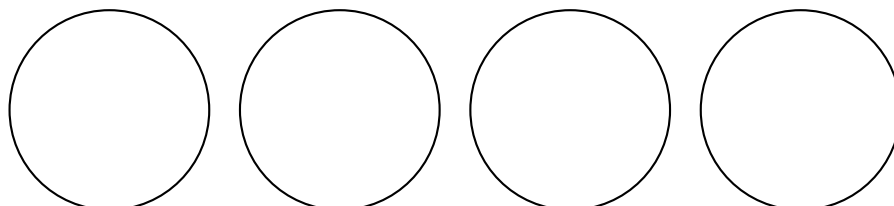


copy of test

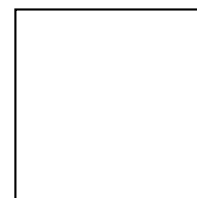
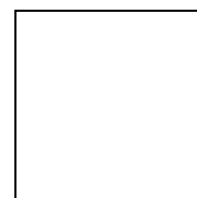
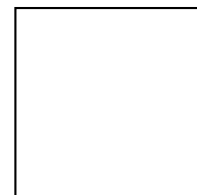


folder

test



score



Problem 6: Find a way to choose your tests so that you always know a four-counter secret after four tests or fewer. Describe it so that someone else could use it.

[illegible][illegible][illegible]This image shows a single sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

Problem 7: Lay out any secret row and any test row, one under the other, and find the score. Then turn over the two counters in the first place, one in each row. What happens to the score? Will that happen whatever the two rows are and whichever place you turn over? Explain.

Problem 8: Explain why one test is not always enough to be sure of a two-counter secret, however you choose it. Then explain why two tests are not always enough to be sure of a three-counter secret, however you choose them.

			score

			score

			score

			score

Problem 9: Can three tests always be enough to be sure of a four-counter secret? Explain.

[illegible][illegible][illegible]This image shows a single sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

Problem 10: Change the three-counter game so that it ends only when a test scores 3. How many tests does the breaker need to be sure of ending the game, whatever the secret is? Answer the same question for four counters, where the game ends when a test scores 4. Explain both answers.

[illegible][illegible]